

Agent Harness Awareness: Verified Operational State as a First-Class Input to Tool-Using Agents

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Final-review proof - formatted from the frozen ToolRoute v1.0 cohort; not yet submitted

Abstract

Tool-using agents are commonly given a task and API descriptions while remaining unaware of the operational state of the harness in which they act. This operational blindness makes a feasible route and a degraded route look identical at the moment the agent chooses. We introduce **Agent Harness Awareness**, a closed-loop design in which compact, fresh, host-verified execution state becomes a first-class input to agent decision making while measurement, scoring, and enforcement remain outside the model. We instantiate the idea in ToolRoute, a controlled routing benchmark with two functionally equivalent tools and seeded changes in observed latency and reliability. Across 216 frozen, hash-finalized direct-provider decisions from GPT-5.6 Sol, Claude Sonnet 5, and Gemini 3.7 Flash, truthful tool-health state reduced mean observable policy regret by 96.9% relative to a neutral same-shape telemetry control (4,252 ms to 131 ms) and increased operational completion from 69.4% to 95.8%. The effect appears in all three model families. These findings establish a foundational systems principle: operational observability can be an active control input, not merely an artifact for human dashboards and post-hoc debugging.

Keywords: tool-using agents; observability; telemetry; agent control; systems; evaluation

1. From operational blindness to active control

An agent can reason carefully about user intent and tool semantics yet still make a poor operational choice. The reason is often not a failure to understand the task. It is that the agent sees a menu of tools but not the current state of the execution harness: which route is slow, failing, stale, rate-limited, resource-constrained, or no longer economical. To the agent, equivalent APIs remain equivalent even when the host already knows that one is degraded.

We call this gap **operational blindness**. Agent Harness Awareness turns verified execution telemetry from a passive observability artifact into an active control input. The interface is intentionally minimal and trust-separated: the host measures raw state, reduces it to a bounded and freshness-labelled snapshot, and retains enforcement authority outside the model. The model may use that state to choose a different high-level action; it cannot edit the measurement, scheduler, oracle, or underlying constraints.

This paper makes two linked contributions. First, it articulates Harness Awareness as a general systems design principle: agents should reason not only over task semantics but also over verified operational state. Second, it provides a rigorous multi-model empirical validation in the tools dimension: verified tool-health state reduces observable policy regret by 96.9% and raises operational completion from 69.4% to 95.8% across three provider/model families.

2. Agent Harness Awareness

2.1 Trust-separated state interface

At a decision time t , let x_t be the task and tool interface, h_t the host state, and $z_t = \varphi(h_t)$ a compact snapshot emitted by the harness. ToolRoute shows why this projection matters: when z_t carries truthful route-relevant monitor facts, agents select actions with substantially lower observable cost than under task-only or neutral same-shape information. A selected, verifiable projection can therefore be a high-leverage operational input without moving measurement or enforcement authority into the model.

The intended contract spans four state families: **Tools** (success, latency, error/circuit state); **Runtime** (deadlines and execution policy); **Hardware** (memory, CPU, disk, processes); and **Economics** (context, quota, rate limits, and cost). ToolRoute empirically evaluates Tools. The remaining families are next empirical instantiations of the same state contract.

2.2 Split planes: deterministic recovery and agent adaptation

The data plane - kernel, proxy, circuit breaker, retry policy, and scheduler - continues to own fast local enforcement and recovery. The agent control plane becomes useful when state changes the strategy: choose a different tool, defer an operation, select a degraded-mode plan, or allocate remaining time and budget differently. ToolRoute isolates this fundamental strategic boundary while preserving deterministic enforcement in the data plane.

3. ToolRoute: a controlled information ablation

3.1 Task and conditions

ToolRoute presents one task, two functionally equivalent tools (tool_alpha, tool_beta), and a wait option. A seeded synthetic host-owned monitor probes each tool three times and records monitor-window success count, latency EWMA, circuit/error state, and snapshot age. The model receives one independent full-checkpoint prompt per decision with no filesystem, shell, network tool, schedule, oracle, or artifact access.

Condition	Model-visible information
A: task only	Task and equivalent tool descriptions; no operational projection.
B: neutral envelope	Same telemetry-shaped envelope, fields, row order, and tool rows as C, with equal neutral values.
C: verified state	Same envelope with fresh host-owned monitor facts for the current tool state.

Condition B is the critical structural control: holding envelope, fields, ordering, and tool rows fixed while neutralizing route-relevant values isolates the contribution of operational state from the presence and shape of the interface. It is not an exact provider-token-matching control; provider input-token counts are retained descriptively.

3.2 External evaluation and auditability

The primary outcome is **observable policy regret**: the excess cost of the selected action under an oracle restricted to canonical monitor facts rendered in C. Successful operational outcome is a secondary binary evaluation. Every provider attempt reserved its artifact directory before invocation and preserved its prompt, sanitized request/response, monitor snapshot, tool event, evaluator result, streams, and finalization hashes. The cohort contains 3 model families x 6 seeds x 4 turns x 3 conditions = 216 independent direct-provider decisions. All records completed and finalized with valid artifact hashes; no request was retried.

3.3 Analysis

The empirical centerpiece is B-to-C: neutral versus truthful operational state in the same interface. We report all decisions in the denominator, Wilson intervals for completion, and raw as well as normalized regret. The primary display reports each model family's B-to-C percentage reduction; raw A/B/C milliseconds remain in the table and appendix. Independent provider generations remain independent; a descriptive bootstrap over 18 model-by-seed environment blocks summarizes variation without treating providers as shared-seed paired draws.

4. Results

Truthful operational state changes the next action substantially. Across 72 decisions per condition, C reduces mean observable policy regret from 4,251.54 ms in B to 131.12 ms - a 4,120.42 ms reduction (96.9%). Completion rises from 50/72 (69.4%; Wilson 95% CI 58.0-79.0%) to 69/72 (95.8%; 95% CI 88.3-98.5%), a 26.4-point increase. Condition A is also materially worse than C (4,826.45 ms regret; 47/72, 65.3% completion).

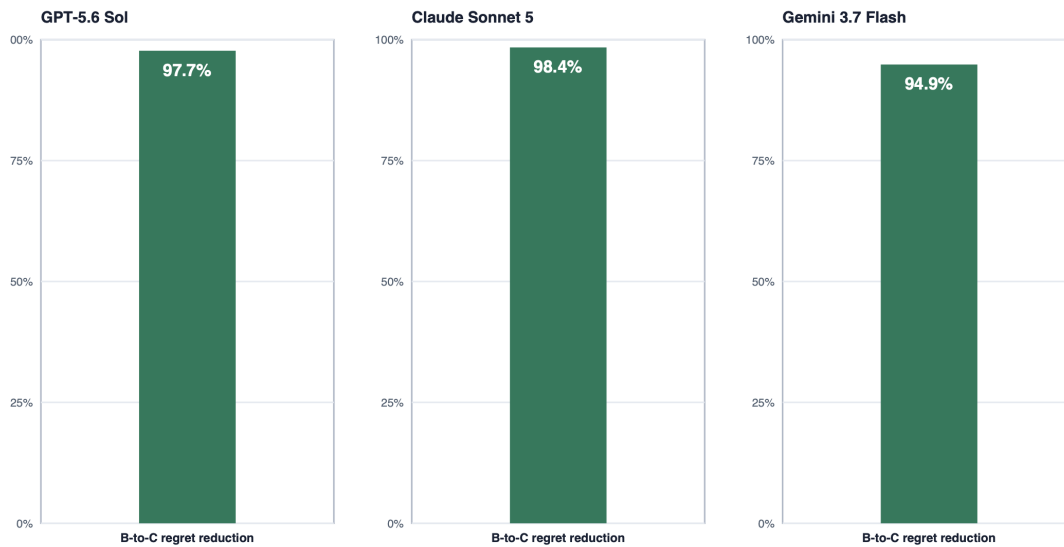


Figure 1. B-to-C reduction in mean observable policy regret. Verified state removes 94.9% to 98.4% of neutral-control regret in every model family; the raw A/B/C milliseconds remain in the table and Appendix Figure A1.

Model	A regret (ms)	B regret (ms)	C regret (ms)	A completion	B completion	C completion
GPT-5.6 Sol	4,723.81	3,307.00	75.50	16/24 (66.7%)	18/24 (75.0%)	23/24 (95.8%)
Claude Sonnet 5	5,031.72	4,723.81	75.50	15/24 (62.5%)	16/24 (66.7%)	23/24 (95.8%)
Gemini 3.7 Flash	4,723.81	4,723.81	242.35	16/24 (66.7%)	16/24 (66.7%)	23/24 (95.8%)

The cross-family repetition matters. The result is not carried by a single provider: C produces a low-regret route in every family, while A and B retain substantial avoidable cost. The three C choices of wait remain in the denominator; they are not filtered away to improve the apparent result.

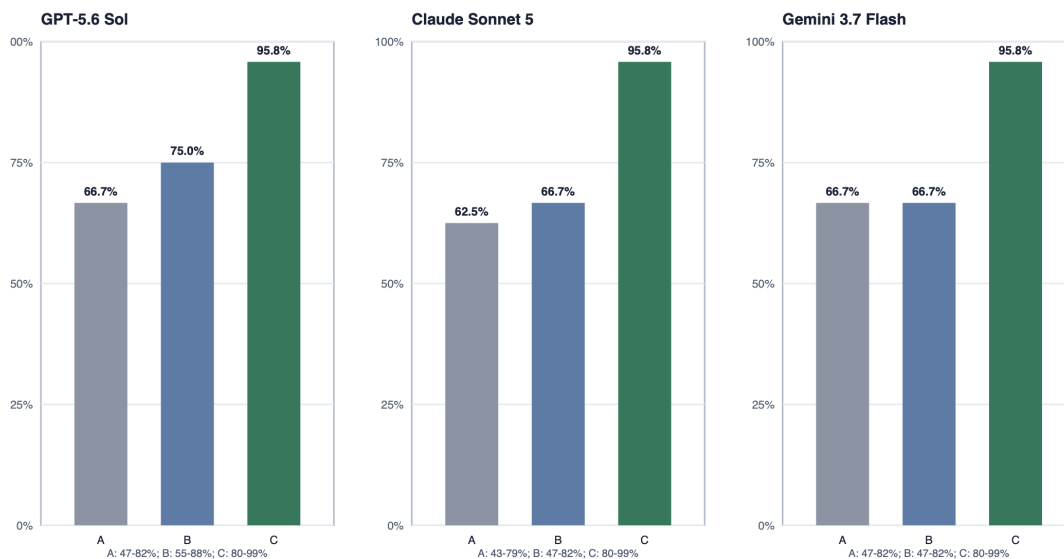


Figure 2. Operational completion by condition and model family. Verified state completes 23 of 24 decisions for each model family.

4.1 What the result establishes

ToolRoute provides strong empirical evidence that verified, current tool-health state is a consequential control input for operational route selection. The B-to-C controlled information ablation holds the telemetry-shaped interface fixed while changing route-relevant values. Model choices are therefore strongly sensitive to structured operational facts, rather than only to the presence of a table-shaped prompt.

5. Scope and research agenda

The experiment isolates one decision type: two-route selection under a synthetic, immediately delivered tool-health monitor. Larger tool catalogs, noisy or delayed monitors, stateful multi-turn histories, changing task semantics, and live service integration are direct next tests. The same interface extends naturally to deadline-aware planning, hardware and concurrency adaptation, economic state, and multi-agent backpressure.

6. Related work

ReAct [1] established the importance of interleaving reasoning with environment actions, while Toolformer [2] studied learned tool use. AgentBench [3] and tau-bench [4] show that agents can fail in evolving, stateful environments. ToolRoute focuses on a different missing observation: verified operational state at route choice. OpenTelemetry [5] provides an established foundation for traces, metrics, and logs; Harness Awareness extends selected, provenance-preserving telemetry toward an agent decision interface. Agent-Native Telemetry [6] is the closest architectural neighbor: it focuses on compact verified telemetry records and access paths, whereas ToolRoute evaluates the downstream behavioral question with a neutral same-shape control.

7. Conclusion

Agent Harness Awareness closes a loop that current tool-using agents often leave open: the harness observes operational state, but the agent chooses as if that state did not exist. In the frozen 216-decision ToolRoute cohort, truthful tool-health state reduced observable policy regret by 96.9% against a neutral same-shape control and increased completion by 26.4 points. These results advance a transition from passive observability to closed-loop agent control: agents can reason over verified execution telemetry while the harness retains authority to measure and enforce it.

Appendix A. Raw policy regret

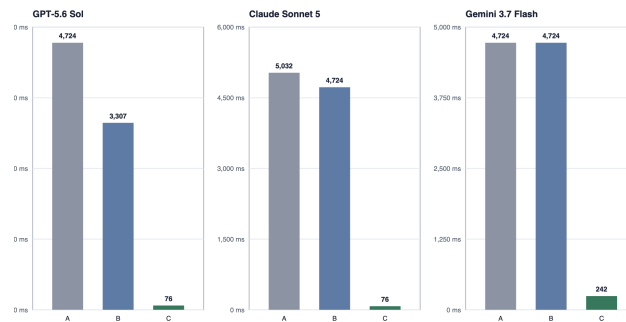


Figure A1. Raw mean observable policy regret in milliseconds. These are the unnormalized values behind Figure 1.

References

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